THE COPPERBELT UNIVERSITY

MCS SCHOOL OF MEDICINE, TEST 3 – OCT 2025, MBS 210 - PHYSIOLOGY

STUDENT NUMBER:.................................... PROGRAMME: ....

TIME ALLOWED: 1.5 hours, TOTAL MARK: 100 marks

Answer SECTION A in the grid sheet provided

SECTION A: CHOOSE THE SINGLE BEST OPTION [80 marks] EACH = 2 MARK

1. Which of the following correctly lists the four layers of the respiratory membrane?
A. Alveolar basement membrane, Type I epithelial cells, Capillary endothelium, Capillary basement membrane
B. Alveolar wall (Type I and Type II epithelial cells), Capillary endothelium, Alveolar basement membrane, Capillary basement membrane
C. Alveolar wall (Type I and Type II epithelial cells), Alveolar basement membrane, Capillary basement membrane, Capillary endothelium
D. Capillary endothelium, Alveolar basement membrane, Type I epithelial cells, Capillary basement membrane
E. Type II epithelial cells, Capillary basement membrane, Capillary endothelium, Alveolar basement membrane

2. Which of the following best explains how sympathetic stimulation affects the heart?
A. Permeability of the S-A node to sodium decreases
B. Permeability of the A-V node to sodium decreases
C. Permeability of the S-A node to potassium increases
D. There is an increased rate of upward drift of the resting membrane potential of the S-A node
E. Permeability of the cardiac muscle to calcium decreases

3. CO_2 is chiefly transported in blood:
A. Bound to Hb
B. Bicarbonate form
C. Plateau form
D. Hyperpolarized state
E. Carbaminohemoglobin

4. Uganda's singer Bobby Whine was giving a concert in Lusaka. The major constituent

of surfactant produced inside lungs is?
A. Palm oil
B. Phosphatidylglycerol
C. Dipalmitoyl phosphatidyl choline
D. Surfactant proteins A & D
E. Permeabilin

5. A globally famous Sr Research Fellow from BBU school of medicine met with pop star DANDY KRAZEE at chishimba falls at a weekend party to celebrate singing and dancing. The maximum amount of gas that can be exhaled after a full inspiration is called:
A. Singing capacity
B. Vital capacity
C. Expiratory capacity
D. Tidal volume
E. Inspiratory capacity

6. The largest volume of gas that can be moved into and out of the lungs in 1 minute by voluntary effort is called:?
A. Muscular ventilation
B. Alveolar capacity
C. Ventilation
D. Minute ventilation
E. Maximum ventilation

7. In a healthy individual with a total lung volume of 6 liters, the amount of oxygen present in the lungs at the end of a tidal expiration is about:
A. 600 ml
B. 2300 ml
C. 500 ml
D. 1200 ml
E. 2400 ml

8. Even with high sophisticated language technology, humans cannot control their own respiratory smooth muscles at will. Intrapleural pressure is positive during?
A. Inspiration
B. Moaning
C. Singing
D. Tidal expiration
E. Forced expiration

9. Kaoma district has a cattle population of around 15000. Due to heavy pollution of air by traffic automobiles, Which of the following shifts the oxyhemoglobin dissociation curve to the left?
A. Increase in 2,3 BPG in RBC

B. High temperature
C. High Carbon monoxide
D. Decrease blood pH
E. Onion rich food

10. Ghaana in West Africa is facing an unprecedented financial crisis, recently. People started looting publicly as well. Normal exhalation in humans, occurs due to this ___________

A. Due to relaxation of expiratory muscles Due
B. Elastic recoil of the lung tissue
C. Depolarization of alveoli
D. Repolarization of bronchial muscles
E. Unknown process

11. A radio jocky at ROSTER FM was delivering a message to the nation. In the systemic capillary:
A. PCO2 is increased from 40 to 45mm Hg
B. PO2 is increased from 95 to 140mm Hg
C. Single-unit muscles are not innervated in lungs
D. Idiopathic
E. PO2 is decreased from 95 to 40mm Hg

12. While attending to an emergency care patient [soldier] who came from conflicted region of TIGRAY mountains, at Ethiopian border with Eritrea, which of the following are used for dealing with acute mountain sickness?
A. 1.9% Nacl
B. 15% Dextrose
C. 100% O2 inhalation
D. 60% O2 inhalation
E. 12% O2 inhalation

13. What enzyme is responsible for breaking down starch into smaller molecules?
A. Dextrinase
B. Salivary amylase
C. Disaccharidase
D. Pancreatic lipase
E. Lactase

14. Which of the following substances is produced in the pancreas to aid carbohydrate digestion?
A. Dextrins
B. Pancreatic amylase
C. Monosaccharides

D. Disaccharidases
E. Salivary amylase

15. The final stage of carbohydrate digestion involves the breakdown of disaccharides into monosaccharides. Which enzymes are responsible for this process?
A. Dextrinase and oligosaccharidases
B. Salivary amylase and pancreatic amylase
C. Disaccharidases
D. Lipases
E. Proteases

16. Which type of transport is involved in sodium-dependent glucose uptake from the intestinal lumen?
A) Primary active transport
B) Secondary active transport
C) Facilitated diffusion
D) Simple diffusion
E) Endocytosis

17. Which statement about sodium-dependent glucose transport (SGLT-1) is TRUE?
A) It depends on insulin for glucose absorption
B) It actively transports glucose against the concentration gradient using ATP directly
C) It also promotes water absorption due to osmotic effects
D) It exhibits transport maximum (TMG), meaning glucose absorption is limited
E) It transports glucose from the blood into the intestinal lumen

18. What transporter moves glucose from the intestinal cells into the bloodstream?
A) SGLT-1
B) GLUT-1
C) GLUT-2
D) GLUT-4
E) Na+/K+ ATPase

19. Which sugar is absorbed by facilitated diffusion in the intestine?
A) Glucose
B) Galactose
C) Fructose
D) Lactose
E) Sucrose

20. Lactose intolerance is primarily caused by a deficiency of which enzyme?
A) Amylase

B) Lactase
C) Maltase
D) Sucrase
E) Lipase

21. How does lactase activity typically change from birth through life?
A) Low at birth and stays low
B) High at birth with progressive decline
C) Constant high throughout life
D) Low at birth and increases with age
E) Fluctuates without pattern

22. Which of the following symptoms is commonly associated with lactose intolerance?
A) Constipation and bloating
B) Increased motility, diarrhea, and cramps
C) Stomach ulcers
D) GERD (acid reflux)
E) Nausea without diarrhea

23. What is the primary treatment recommendation for lactose intolerance?
A) Increase lactose consumption gradually
B) Use insulin injections
C) Elimination of lactose from the diet (milk and milk products)
D) Take antibiotics
E) Drink carbonated beverages frequently

24. Which of the following is considered a choleretic agent?
A. Insulin
B. Gastrin
C. Bile salts
D. Trypsin
E. Amylase

25. Which of the following substances acts as a cholagogue agent that increases the flow of bile?
A. Glucose
B. Calcium
C. Pepsin
D. Lipase
E. Insulin

26. Cholic acid conjugated with which of the following amino acids forms taurocholic acid?

A. Glycine
B. Taurine
C. Glutamine
D. Alanine
E. Lysine

27. The sodium (Na) and potassium (K) salts of taurocholic and glycocholic acids are collectively known as:
A. Bile pigments
B. Bile salts
C. Fatty acids
D. Cholesterol
E. Phospholipids

28. Which of the following neural mechanisms increases the production and flow of bile?
A) Sympathetic nerve stimulation
B) Stimulation of the glossopharyngeal nerve
C) Vagal nerve stimulation
D) Inhibition of the parasympathetic nervous system
E) Activation of the splanchnic nerves

29. The hormone that stimulates bile production is:
A. Cholecystokinin (CCK-PZ)
B. Secretin
C. Insulin
D. Gastrin
E. Glucagon

30. What is the specific action of Cholecystokinin-Pancreozymin (CCK-PZ) regarding bile, as stated in the text?

A) It inhibits the production of bile by liver cells.
B) It stimulates the absorption of bile salts in the ileum.
C) It causes the relaxation of the sphincter of Oddi.
D) It decreases the flow of bile from the gallbladder.
E) It increases the flow of bile from the gallbladder.

31. Which of the following substances or measures is not explicitly listed as a test used to assess the secretory functions of bile/liver in the provided text?

A) Serum Bilirubin
B) Faecal stercobilinogen
C) Urine bile salts

D) Urine urobilinogen
E) Serum Albumin

32. The cascade of events at high altitude begins with hypoxic hypoxia which ultimately leads to "physiological derangements." What is the immediate preceding event that directly results from the hypoxic hypoxia and causes these derangements?

A) Increased red blood cell production (polycythemia)
B) Hyperventilation and respiratory alkalosis
C) Reduced heart rate (bradycardia)
D) Suffering of tissue oxygenation
E) Decreased partial pressure of oxygen in the atmosphere

33. What term is used to describe the long-term process by which the body adapts to remain at high altitudes and cope with the chronic condition of low partial pressure of oxygen (PO2)?

A) Acute Mountain Sickness (AMS)
B) Hyperventilation Syndrome
C) Ischemic Hypoxia
D) Acclimatization
E) Ventilatory Depression

34. For individuals, such as those living in the Andes and Himalayan mountains, what physiological adaptation begins in infancy and results in a greatly increased chest size, as a response to the low PO2 environment?

A) Increased oxygen-carrying capacity due to a higher concentration of myoglobin in muscles.
B) Long-term developmental changes resulting from childhood exposure to chronic hypoxia.
C) Rapid, short-term hyperventilation to increase blood oxygen saturation.
D) A pathological condition known as Chronic Mountain Sickness (CMS).
E) A genetic mutation that prevents hemoglobin from releasing oxygen to the tissues

35. Acute Mountain Sickness (AMS) is a condition that occurs primarily in a small number of lowlanders like Amirunniya, who rapidly ascend to high altitudes. How long after their ascent does AMS typically begin?
A) Immediately upon reaching the high altitude
B) From a few hours up to 2 days
C) Approximately one week
D) Only after remaining at high altitude for several months
E) Within 3 to 7 days

36. Deboraah who has resided at high altitudes for a long time develops Chronic Mountain Sickness (CMS). Which of the following is the key physiological characteristic associated with this condition?
A) A sharp, short-term decrease in the number of circulating white blood cells.
B) The failure of the body to initiate hyperventilation in response to low PO2.
C) An exceptional increase in the number and total mass of red blood cells.
D) Severe dehydration and electrolyte imbalance due to excessive urination.
E) An immediate onset of dizziness and nausea within hours of arriving at altitude.

37. The "Release Phenomena" observed during the initial accommodation phase at high altitude is characterized by the depression of the Central Nervous System (CNS). What common effect does this phenomenon resemble?
A) The physical symptoms of a fever, including chills and muscle aches.
B) The stimulating effects of high-dose caffeine consumption.
C) The disorientation and cognitive impairment similar to the effect of alcohol.
D) The acute onset of pulmonary edema and fluid in the lungs.
E) The calming, sedative action of antianxiety medication.

38. Prolonged hyperventilation causes a phenomenon known as "CO2 wash-out." This rapid decrease in the partial pressure of carbon dioxide (PCO2) in the blood immediately results in which primary acid-base disturbance?
A) Metabolic Acidosis
B) Metabolic Alkalosis
C) Respiratory Acidosis
D) Respiratory Alkalosis
E) Mixed Acidosis

39. Which organ system provides compensation for the respiratory alkalosis caused by prolonged hyperventilation, and what is the resulting key change?
A) Central Nervous System, causing sleepiness.
B) Pulmonary System, decreasing respiratory rate.
C) Hepatic System, decreasing plasma protein production.
D) Renal System, resulting in alkaline urine.
E) Endocrine System, increasing parathyroid hormone.

40. Which statement best describes the critical phase of Cheyne-Stokes respiration?
A) Breathing is continuously deep and rapid (hyperpnea).
B) Breathing depth remains constant but is very shallow.
C) Respiration progresses from shallow to deep, then pauses completely.
D) Breathing arrests for a few seconds, often causing the person to wake suddenly feeling suffocated.
E) Breathing is constantly irregular without any predictable cycle

THE END